

1. Given the equation $x^n + y^n = z^n$ for x, y, z and n positive integers.
 - (a) (5 points) For what values of n is the statement in the previous question true?
 - (b) (2 $\frac{1}{2}$ points) For $n = 2$ there's a theorem with a special name. What's that name?
 - (c) (2 $\frac{1}{2}$ points) What famous mathematician had an elegant proof for this theorem but there was not enough space in the margin to write it down?

Total for Question 1: 10

2. (20 points) Compute

$$\int_0^\infty \frac{\sin(x)}{x}$$

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3. (30 points (bonus)) Prove that the real part of all non-trivial zeros of the function $\zeta(z)$ is $\frac{1}{2}$

Question:	1	2	3	??	??	??	??	Total
Points:	10	10	0	????	????	????	????	20
Bonus Points:	0	0	30	????	????	????	????	30
Score:								

Here we'll use `\pointsinmargin`

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Now we'll use `\pointsinrightmargin`

2. Given the equation $x^n + y^n = z^n$ for x, y, z and n positive integers.

- (a) For what values of n is the statement in the previous question true? (5)
- (b) For $n = 2$ there's a theorem with a special name. What's that name? (2 $\frac{1}{2}$)
- (c) What famous mathematician had an elegant proof for this theorem but there was not enough space in the margin to write it down? integers. (2 $\frac{1}{2}$)